

GENERAL NOTES:

1- PIPE PASSING THROUGH WALL, FLOOR OR ROOF SHALL BE PROTECTED WITH SLEEVES.

2- CLEAN OUT OR ACCESS DOOR SHALL BE PROVIDED AT THE BASE OF ALL DRAINAGE STACK AND AT EACH FLOOR LEVEL.

3- DRAINAGE STACKS SHALL BE CONNECTED TO HORIZONTAL FIXTURE DRAIN PIPES BY 90° SWEEP TEE OR Y AND 45° BEND.

4- DRAINAGE STACKS SHALL BE CONNECTED TO UNDERGROUND DRAIN PIPE BY EITHER 90° LARGE/LONG RADIUS BEND OR TWO 45° BENDS.

5- PIPE SIZES ARE IN mm.

6- PLUMBING SERVICES, SHALL BE COORDINATED WITH OTHER SERVICES SUCH AS ELECTRICAL, HVAC ETC. TO AVOID CONFLIT OR INTERFERENCE WITH THEM.

7- SIZE OF FLOOR DRAIN UNLESS OTHERWISE STATED SHALL BE SAME AS THE SIZE OF OUTLET DRAIN PIPE.

8- WHEREVER FLOOR DRAIN/TRAP IS SHOWN, SLOPE FLOOR TOWARDS DRAIN/TRAP.

9- HORIZONTAL DRAINAGE PIPING OF 82mm DIAMETER & LESS SHALL BE INSTALLED WITH A SLOPE OF NOT LESS THAN 1% AND LARGER ONES WITH A SLOE OF NOT LESS THAN 0.5%.

10- EXTENSION OF VENT PIPES THROUGH A ROOF SHALL BE TERMINATED AT LEAST 150mm ABOVE FINISHED ROOF LEVEL. EACH VENT TERMINAL SHALL BE MADE WATER TIGHT WITH THE ROOF BY PROPER FLASHING.

11- THE "VERTICAL DISTANCE" FROM FIXTURE OUTLET TO THE TRAP WEIR SHALL PREFERABLY BE NOT MORE THAN 300mm. HOWEVER, THIS "VERTICAL DISTANCE" FROM THE FIXTURE OUTLET TO THE TRAP WEIR SHALL NOT EXCEED 600mm.

12- WHERE THE CHANGE OF DIRECTION IS 45° OR LESS, IT IS NOT NECESSARY TO PROVIDE A CLEAN OUT. BUT WHEN THE CHANGE IN DIRECTION IS MORE THAN 45°, A CLEAN OUT SHOULD BE REQUIRED.

13- CLEANOUTS SHALL BE SO INSTALLED THAT THERE IS A CLEARANCE OF NOT LESS THAN 450mm FOR THE PURPOSE OF RODDING.

14- ALL TRAPS AND FLOOR DRAINS SHALL HAVE A WATER SEAL NOT LESS THAN 50mm.

15- PROVIDE TEMPERATURE RELIEF VALVES OR ENERGY SHUT OFF DEVICES FOR ALL EQUIPMENT USED FOR THE HEATING OR STORAGE OF HOT WATER. THE RELIEVING CAPACITY OF A TEMPERATURE-RELIEF VALVE SHOULD NOT BE LESS THAN ONE GALLON OF WATER PER HOUR FOR EACH 1250 BTU. OF HEATER CAPACITY PER HOUR.

16- (EXTERNAL SERVICES) THE BOTTOM OF WATER-SERVICES PIPE, AT ALL POINTS, SHALL BE AT LEAST 12 INCHES ABOVE THE TAP OF THE SEWER LINE AT ITS HIGHEST POINT.

17- NO "T" CONNECTION IS ALLOWED IN SOIL AND WASTE PIPE LINES.

18- EVERY WATER SUPPLY FIXTURE INCLUDES A GATE VALVE

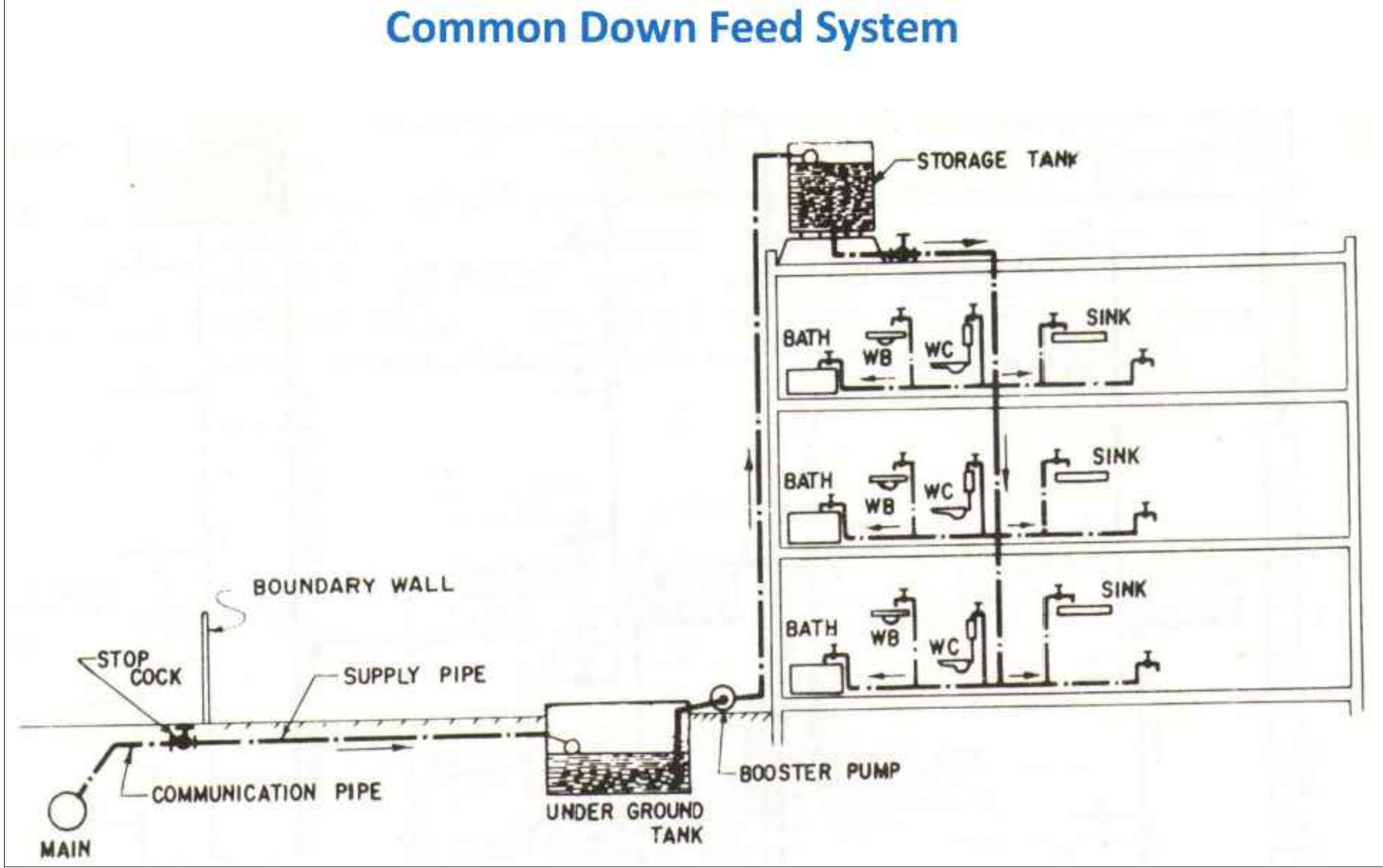
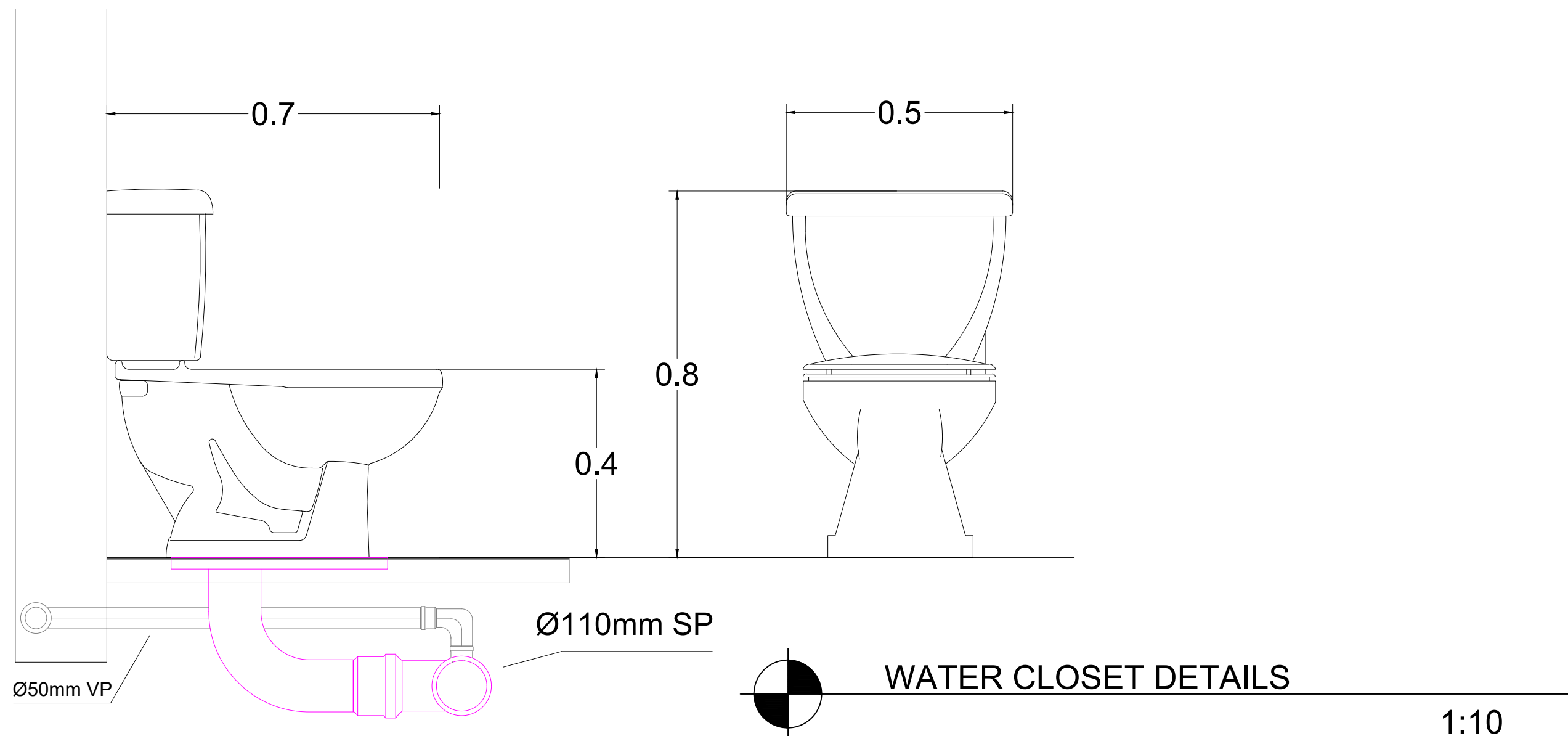
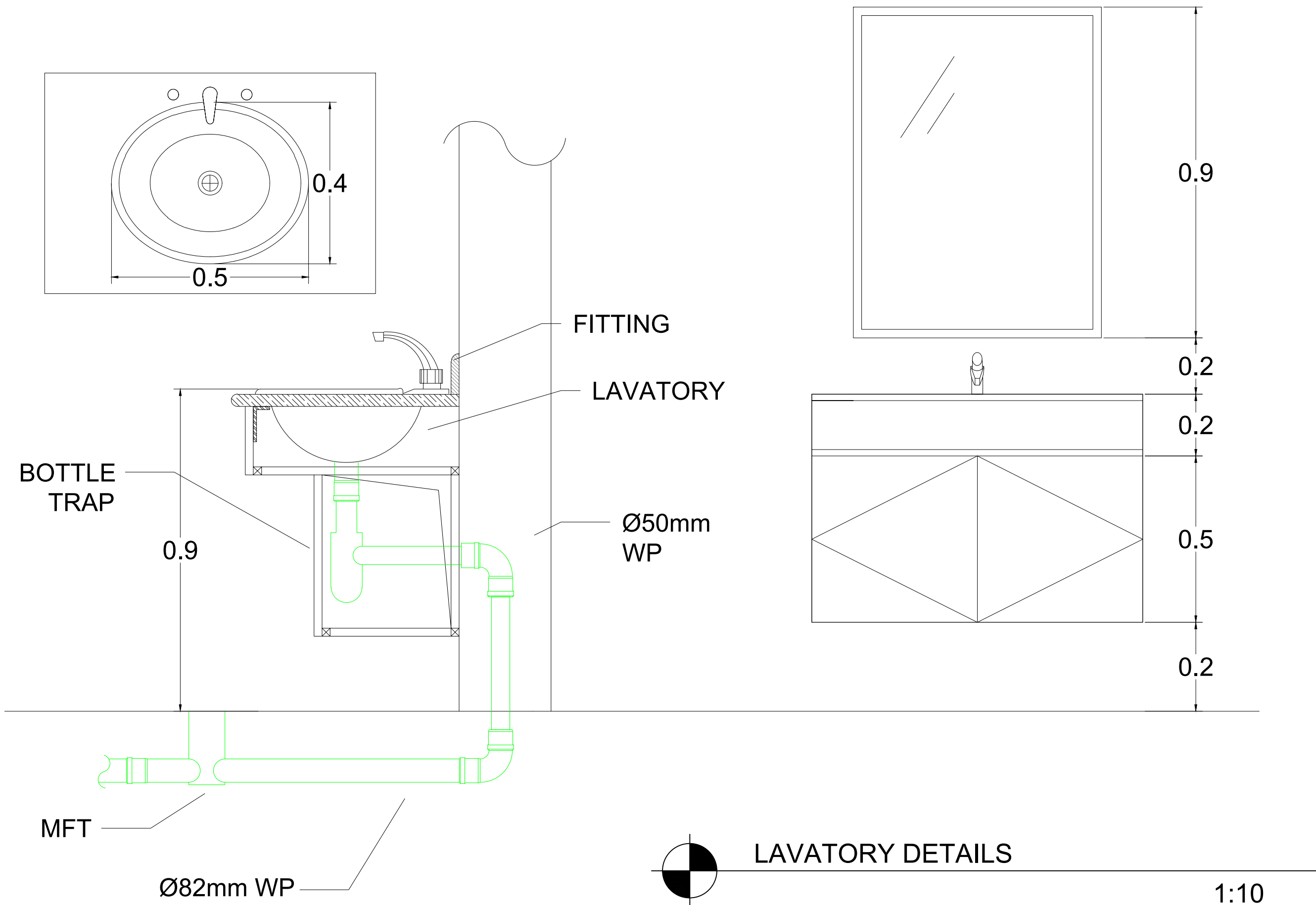
LEGEND	
SYM.	DESCRIPTION
MFT	- MULTI FLOOR TRAP
WP	- WASTE PIPE
WC	- WESTERN WATER CLOSET
LAV	- LAVATORY
VP	- VENT PIPE
AF/C	- ABOVE FALSE CEILING
GV	- GATE VALVE
VS	- VENT STACK
SP	- SEWER PIPE
WP	- WASTE WATER PIPE

Thumb Rule Pipe Diameters Water Supply

No. of fixtures	Pipe Diameter
1 Fixture	15 mm
2-3 Fixtures	20 mm
4-6 Fixtures	25 mm
6-10 Fixtures	32 mm
11-15 Fixtures	40 mm

Available Pipe Diameters

Plastic Pipes (External Diameters)			Metallic Pipes (Internal Diameters)	
Water Supply	Drainage	Equivalent in Inches	Equivalent in Inches	
15mm		3/8"	15mm	3/4"
20mm		3/4"	20mm	3/4"
25mm		1"	25mm	1"
32mm		1 1/4"	32mm	1 1/4"
40mm	40mm	1 1/2"	40mm	1 1/2"
50mm	50mm	2"	50mm	2"
63mm	63mm	2 1/2"	65mm	2 1/2"
82mm	82mm	3"	80mm	3"
110mm	110mm	4"	100mm	4"
160mm	160mm	6"	150mm	6"
200mm	200mm	8"	200mm	8"

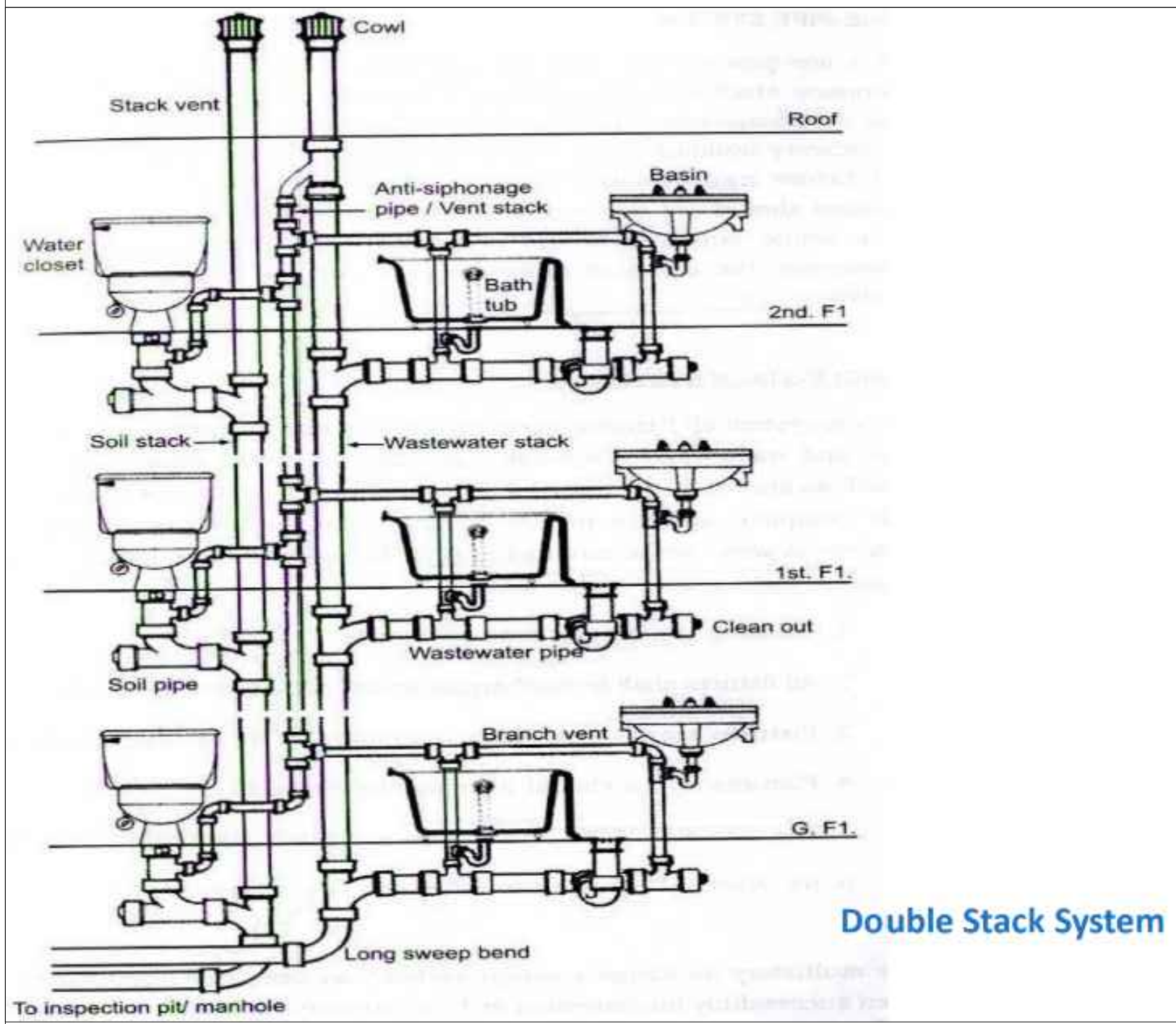


COMMON DOWN FEED SYSTEM

USED IN PROJECT

-The system in which the fixtures and fittings are connected with the supply pipe having flow in downward direction. It comes in process when a building has over head water storage tank. **Residential buildings are usually designed as down-feed system.**

- Simple design**, with water supplied from a higher point and distributed downwards.
- Gravity-assisted drainage**, **reducing the need for pumps and facilitating easier installation and maintenance.**
- Potential for diminished water pressure on higher floors.
- Dependence on gravity for water distribution, limiting suitability for buildings with irregular layouts.



DOUBLE STACK SYSTEM

USED IN PROJECT

Double Stack System:

-In a double stack system, there are two separate vertical stack pipes: one for soil waste and another for wastewater (also known as a foul stack and a waste stack).

-The foul stack collects sewage and wastewater from toilets and other soil fixtures, while the waste stack collects wastewater from sinks, showers, and other fixtures.

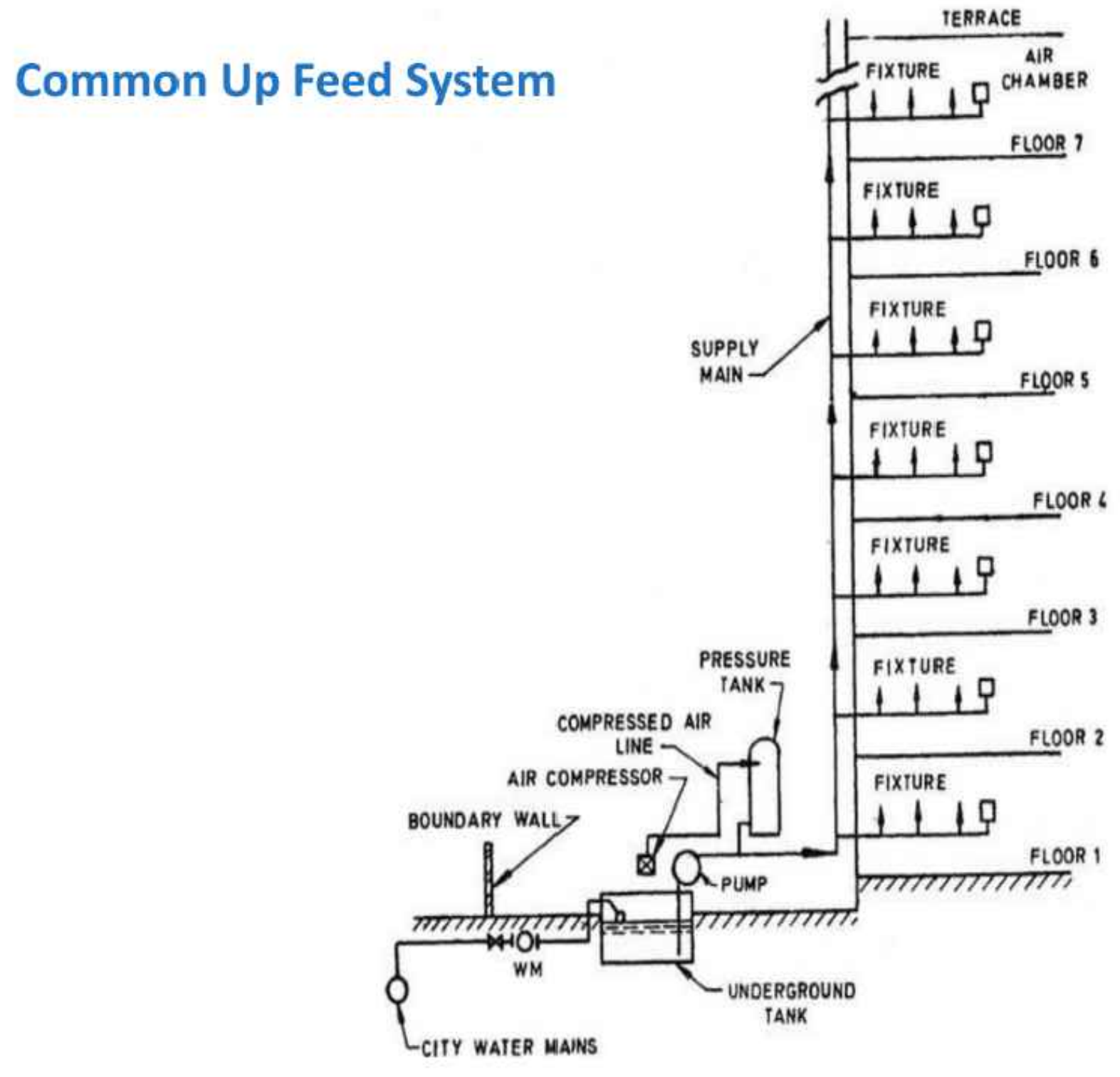
-**Double stack systems provide separate drainage for soil waste and wastewater, reducing the risk of blockages and improving drainage efficiency.**

-**They can handle higher volumes of wastewater and are less prone to pressure fluctuations compared to single stack systems.**

-However, double stack systems are more complex and require additional piping and fittings compared to single stack systems.

-Installation and maintenance costs for double stack systems are typically higher than for single stack systems.

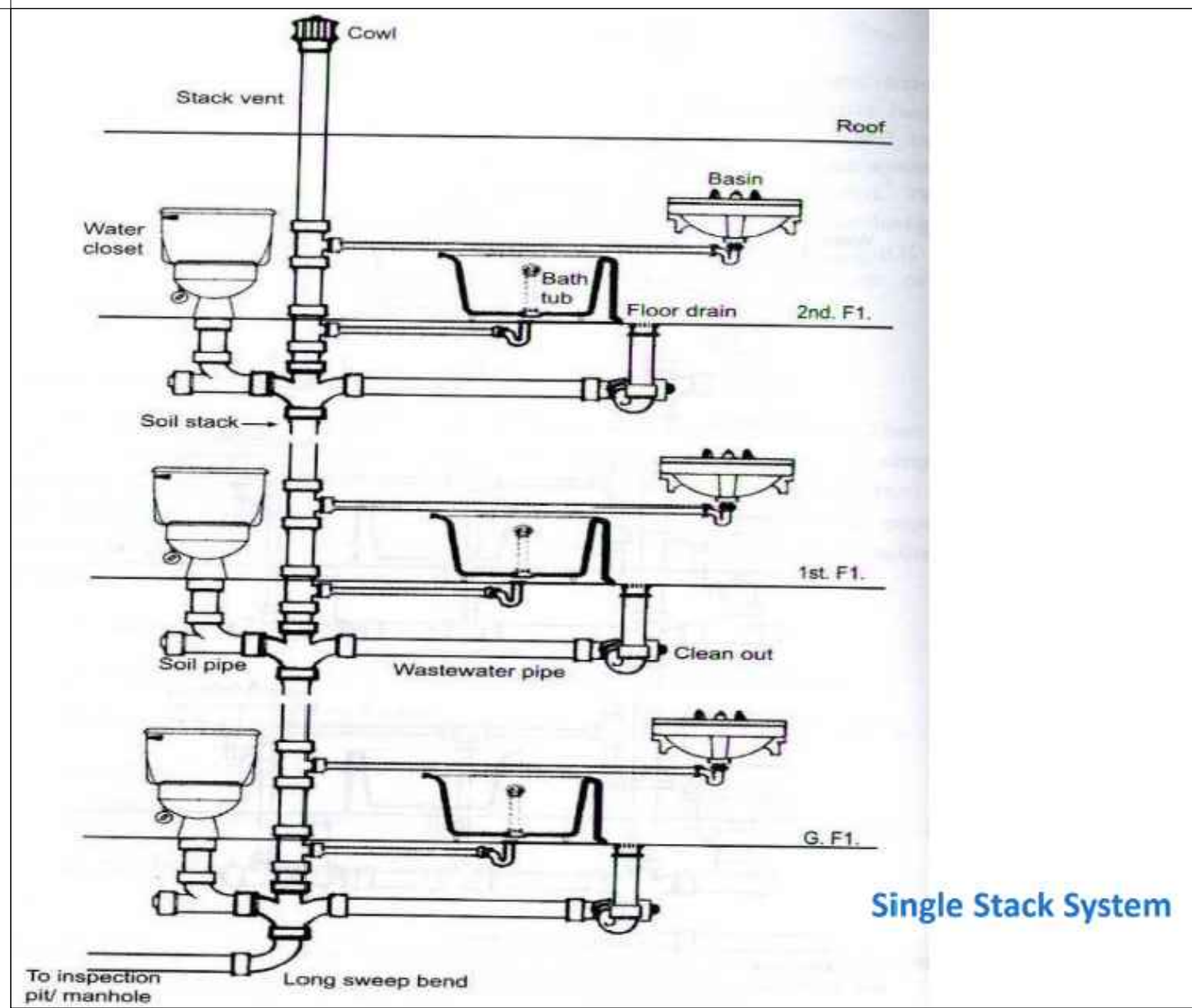
Common Up Feed System



COMMON UP FEED SYSTEM NOT USED IN PROJECT

-The system in which the fixtures and fittings are connected with the supply pipe having flow in upward direction. This type of system generally does not have a water storage tank above the building.

- Water supplied from a lower point and distributed upwards, ensuring consistent water pressure throughout the building.
- Reduced risk of contamination and reliance on gravity during power outages.
- Complexity in design and installation, potentially higher initial costs, and increased risk of leaks.
- Space-efficient and suitable for buildings with limited footprint or layout constraints.



SINGLE STACK SYSTEM NOT USED IN PROJECT

Single Stack System:

-In a single stack system, all wastewater and sewage from various fixtures on different floors are collected into a single vertical stack pipe.

-The single stack runs through the entire height of the building, typically located within a service shaft or riser.

-Each floor's wastewater is discharged into the stack at its respective level, and vents are installed at strategic points to prevent air pressure issues and ensure proper drainage.

-Single stack systems are simpler in design and require fewer pipes and fittings compared to double stack systems.

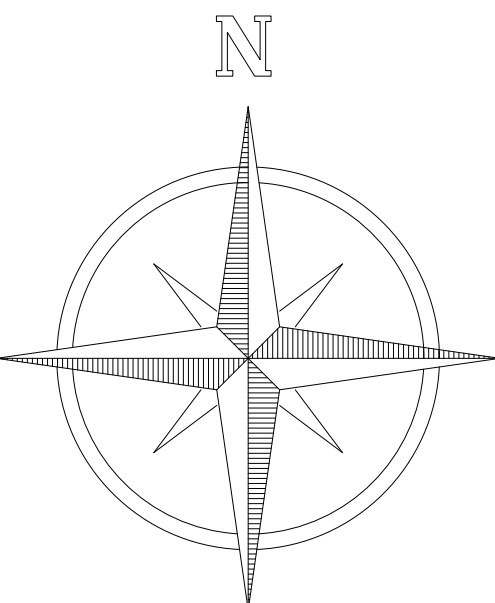
-They are generally more cost-effective to install and maintain.

-However, single stack systems may experience greater pressure fluctuations and the potential for blockages due to the accumulation of waste from multiple floors in a single stack.



جامعة الإمام عبد الرحمن بن فيصل
IMAM ABDULRAHMAN BIN FAISAL UNIVERSITY

COLLEGE OF ARCH. & PLAN
DEPT. OF BUILDING ENGINEERING



ALL UNITS IN METER



كود البناء السعودي
Saudi Building Code



ICC
INTERNATIONAL CODE COUNCIL®

Saudi Plumbing International
Code SBC-701 Plumbing Code
(IPC)

- SOIL WATER PIPE
- WASTE WATER PIPE
- VENTILATION PIPE
- COLD WATER SUPPLY
- HOT WATER SUPPLY

Design Studio (6), Technical
Project, 2023/2024

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PROJECT:
STUDENT DORM

DRAWING NAME:
PLUMBING SYSTEM
SELECTION & DETAILS

SCALE:
1:20

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